



Unveiling correlations between financial variables and topological metrics of trading networks: Evidence from a stock and its warrant



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HIGHLIGHTS

- We construct and investigate time series of security trading networks.
- Correlation relationships are uncovered between trading network metrics and financial variables.
- The structure of trading networks contains rich information about the dynamics of securities.

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ABSTRACT

Traders develop and adopt different trading strategies attempting to maximize their profits in financial markets. These trading strategies not only result in specific topological structures in trading networks, which connect the traders with the pairwise buy–sell relationships, but also have potential impacts on market dynamics. Here, we present a detailed analysis on how the market behaviors are correlated with the structures of traders in trading networks based on audit trail data for the Baosteel stock and its warrant at the transaction level from 22 August 2005 to 23 August 2006. In our investigation, we divide each trade day into 48 rolling time windows with a length of 5 min, construct a trading network within each window, and obtain a time series of over 11,600 trading networks. We find that there are strongly simultaneous correlations between the topological metrics (including network centralization, assortative index, and average path length) of trading networks that characterize the patterns of order execution and the financial variables (including return, volatility, intertrade duration, and trading volume) for the stock and its warrant. Our analysis may shed new lights on how the microscopic interactions between elements within complex system affect the system's performance.

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1. Introduction

Trading networks, in which the nodes represent the traders and the edges stand for the trading relationships, are much used to model the buy–sell interactions among traders in economic systems. Due to the availability of audit trail data at the transaction level, many effects have been put into the understanding of the topological characteristics of trading networks. Kyriakopoulos et al. find that the Austrian money flow trading network is disassortative and also exhibit low correlation between node degrees and transaction volume [1]. Tseng et al. also discover that the trading network in a web-based experimental prediction market is scale-free and disassortative. Jiang and Zhou reveal that the daily trading networks in Shenzhen stock market exhibit patterns of power-law degree distributions and disassortative architectures [2]. Tseng et al. uncover a power-law correlation between the average degree and the size of placed orders [3,4]. Wang et al. find that the trading networks in the Shanghai Futures Market exhibit similar features such as scale-free, small-world effect, hierarchical organization, and power-law betweenness distribution [5].

Trading networks provide an overall perspective to describe the detailed transactions between pairwise traders, which can be applied to find clues and propose approaches to track and detect the abnormal trades implemented by manipulators in financial markets. Kyriakopoulos et al. report that the random matrix analysis is able to identify accounts with financial misconduct [1]. Tumminello et al. identify clusters of traders with a very high degree of synchronization in trading [6], which probably is also related to price manipulation. By investigating the trading networks of manipulated and non-manipulated stocks in the Shanghai stock market, Sun et al. observe significant differences in the topological properties between manipulated and non-manipulated stocks [7,8]. Jiang et al. find that the abnormal trading motifs (self-loop, two-node loop, and two-node multiple arcs) in trading networks have connections with turbulent market dynamics and can be adopted to develop novel tools for the detection of trade-based price manipulations [9]. Sun et al. also find that trading networks predict stock price movements at the daily level [10].

Trading networks are formed based on the transactions triggered by traders with different trading strategies, which provide two perspectives to understand the trading behaviors of traders. On the one hand, the successful strategies can be followed and copied by other traders and the traders with correlated strategies will lead to similarity both in returns and positions in trading networks. Cohen et al. report that traders' returns are correlated with their positions occupied in the trading networks [11]. They also find that the market shocks at the individual level can be widely transmitted and greatly amplified through the links of trading networks. On the other hand, the specific trading strategies of traders may lead to specific local structures in trading networks and also have some influences on market behaviors. Adamic et al. explore the trading networks constructed from the September 2009 E-mini S&P 500 futures contracts and found that there are contemporaneously correlated relationships between network metrics (centralization, assortative index and average path length) and financial variables (return, volatility, intertrade duration, and trading volume) [12].

In this paper, we follow Ref. [12] to investigate correlations between financial variables and topological metrics of trading networks of a liquid Chinese stock and its warrant. The remaining part of this paper is organized as follows. In Section 2, we briefly describe the data sets we adopt. Sections 3 and 4 define respectively the topological metrics of trading networks and the financial variables investigated in this work. The correlations between financial variables and topological metrics of trading networks are presented in Section 5. We summarize our findings in Section 6.

2. Data sets

We employ a transaction-level database, which contains the actual trades completed on the Shanghai Stock Exchange for stock Baosteel and its warrant, to uncover the potential connection between the equities' behaviors and the topological structures of trading networks. The Shanghai securities market is a pure order-driven market and the observed trades are triggered by matching the market orders or executable limit orders with the orders sitting on the opposite side of limit order book, which is a waiting queue of limit orders sorted by price and time priority. There are totally 3,779,538 transactions for stock Baosteel and 25,483,344 transactions for its warrant during our sampling period from 22 August 2005 to 23 August 2006, covering 243 trading days. Each transaction contains the following information: a unique transaction ID, two unique trader IDs, the trading time, the trade size, the transaction price, and an indicator of buy or sell. We note that the stock (warrant) is traded according to the $T + 1$ ($T + 0$) rule.

3. Trading network metrics

In this section, we describe briefly the construction of trading networks and show several examples of randomly chosen trading networks for the stock and its warrant. Then, five topological metrics based on network centralization, assortative index, and average path length are adopted to quantitatively characterize the structure of each trading network. We further provide explicit definitions of the network metrics, illustrate these time series, and investigate their basic statistics.

3.1. Construction of trading networks

For both securities, we divide each trading day into 48 time windows with an interval of 5 min. In each window, the transaction records are translated into one trading network, in which the nodes represent the traders and the links weighted

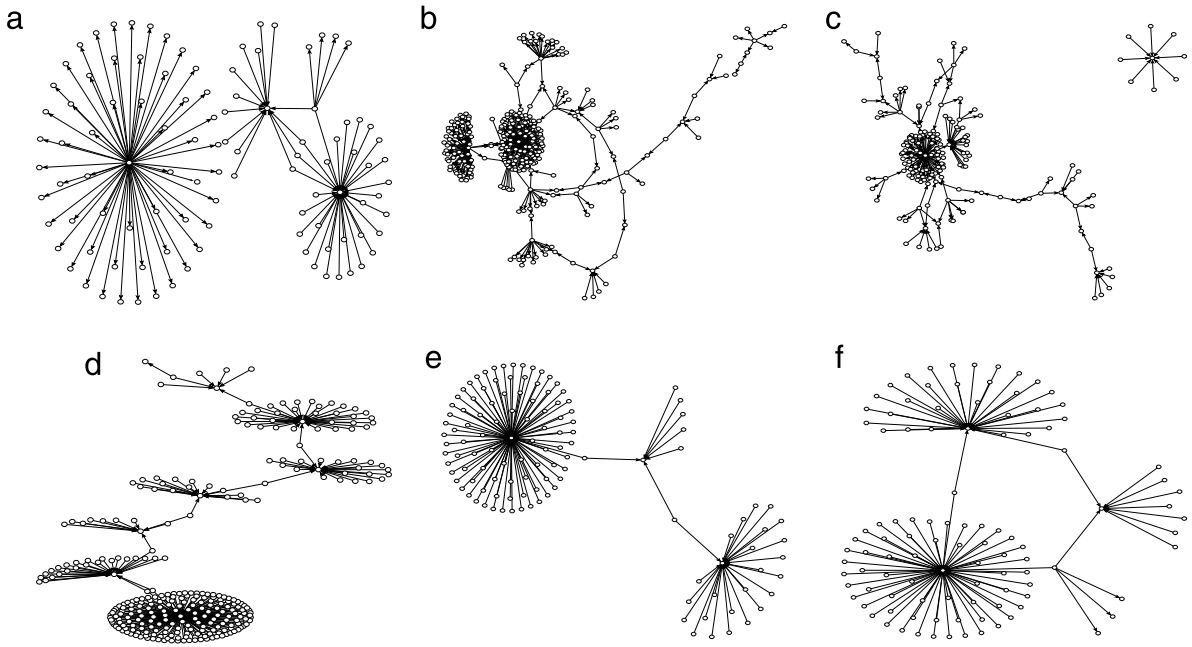


Fig. 1. Examples of randomly chosen trading networks in 5-min windows. Plots (a)–(c) are for the stock, while plots (d)–(f) are for the warrant. The arrows direct from sellers to buyers. The most striking feature of these networks is the star-like local structure.

by the trade sizes are connected from the sellers to the buyers [2]. Denote n the number of traders in a window and m the number of edges or transactions. We obtain that $n = 154 \pm 170$ and $m = 161 \pm 185$ for the stock and $n = 990 \pm 1139$ and $m = 1093 \pm 1299$ for the warrant. The large standard deviations of the four quantities show that the structure of trading networks vary remarkably along time. The fact that the warrant trading networks have more nodes and edges than the stock trading networks implies that the trading activity of the warrant is higher than the stock during the time period under investigation.

In Fig. 1, we show three randomly chosen trading networks in 5-min windows for each security. The trading networks can be connected or isolated. Regardless of the link direction, most trading networks are trees. However, there are also trading networks with loops, some of which are probably caused by price manipulators [9]. The most striking feature of these networks is a star-like local structure. For instance, the trading network in Fig. 1(c) have seven “stars” with the central nodes being sources (big sellers) or sinks (big buyers) and these stars are connected by one or two nodes. Certainly, there are also segments of chains, especially for stock trading networks, whose chain lengths are usually not too long.

3.2. Network centrality

The unweighted network centralization C_k is defined as the difference between the out-degree centralization C_k^{out} and the in-degree centralization C_k^{in} , such that $C_k = C_k^{\text{in}} - C_k^{\text{out}}$. Here we use the degree to measure the node centrality. To avoid possible confusion, we note that the subscript k stands for the degree, not the index of a node. This notation also applies to the variable e_k defined below. The in-degree centralization can be estimated through a well-known definition [13]:

$$C_k^{\text{in}} = \frac{\sum_{i=1}^n (k_{\max}^{\text{in}} - k_i^{\text{in}})}{(n-1)^2}, \quad (1)$$

where k_i^{in} is the in-degree of node i , $k_{\max}^{\text{in}}$ is the maximal in-degree and n is the number of nodes. The denominator $(n-1)^2$ in Eq. (1) represents the maximum possible sum of differences in point centrality for a directed graph of n nodes. Hence, C_k^{in} varies between 0 and 1. By replacing the in-degree with the out-degree, we can obtain the out-degree centralization,

$$C_k^{\text{out}} = \frac{\sum_{i=1}^n (k_{\max}^{\text{out}} - k_i^{\text{out}})}{(n-1)^2}. \quad (2)$$

C_k^{out} also locates in the range of $[0, 1]$.

The measure C_k falls in the interval $[-1, 1]$ [14]. If a network has a star-like core–periphery structure in which all the arrows are from the peripheral nodes to the core node such that the core node acts as a sink, the in-degree and the

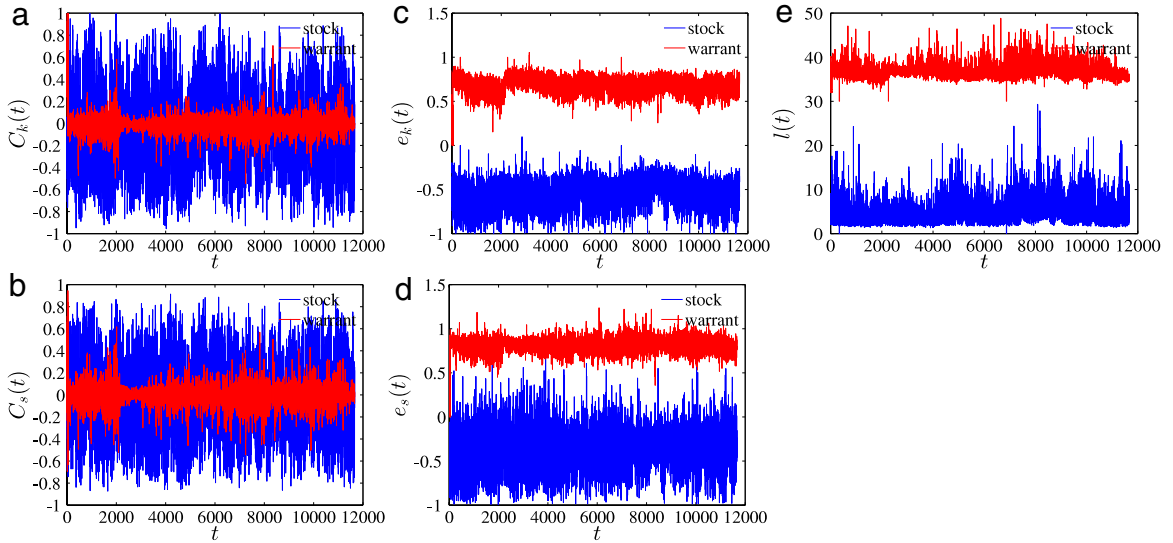


Fig. 2. (Color online) Time series of the network metrics measure in 5-min rolling windows for the Baosteel stock and its warrant: (a) unweighted centralization C_k , (b) weighted centralization C_s , (c) unweighted assortative index e_k , (d) weighted assortative index e_s , and (e) average path length l . To have better visibility, the time series of e_k and e_s for the warrant have been moved upwards by 1 and the time series of l for the warrant has been moved up by 30.

out-degree of the sink are respectively $k_{\max}^{\text{in}} = n - 1$ and 0, while the in-degree and out-degree of the peripheral nodes are 0 and $k_{\max}^{\text{out}} = 1$. We have $C_k^{\text{in}} = 1$ and $C_k^{\text{out}} = 1/(n - 1)^2$, which leads to $C_k \approx 1$ when n is large. If a network has a star-like core–periphery structure in which all the arrows are from the core node to the peripheral nodes and the core node acts as a source, the in-degree and the out-degree of the source are respectively 0 and $k_{\max}^{\text{out}} = n - 1$, while the in-degree and out-degree of the peripheral nodes are $k_{\max}^{\text{in}} = 1$ and 0. We have $C_k^{\text{in}} = 1/(n - 1)^2$ and $C_k^{\text{out}} = 1$, which leads to $C_k \approx -1$ when n is large. These two cases are not usual and can occur when the time window is small. The cases that $C_k = \pm 1$ indicate the presence of a dominating buyer and a dominating seller. The two time series of $C_k(t)$ for the two securities are illustrated in Fig. 2(a). Both unweighted centralization measures are centered around zero. However, the warrant's C_k has much smaller fluctuations than the stock.

The weighted network centralization C_s is extended from the unweighted network centralization C_k , in which the node centrality is measured by its node strength. Similar to C_k^{in} , we can write the in-strength centralization as

$$C_s^{\text{in}} = \frac{\sum_{i=1}^n (s_{\max}^{\text{in}} - s_i^{\text{in}})}{(n - 1) \sum_i s_i^{\text{in}}}, \quad (3)$$

where s_i^{in} is the in-strength of node i , $s_{\max}^{\text{in}}$ is the maximum in-strength, and n is the number of nodes. The out-strength centralization can be also written as

$$C_s^{\text{out}} = \frac{\sum_{i=1}^n (s_{\max}^{\text{out}} - s_i^{\text{out}})}{(n - 1) \sum_i s_i^{\text{out}}}. \quad (4)$$

Obviously, both C_s^{in} and C_s^{out} will vary between 0 and 1. The weighted network centralization $C_s = C_s^{\text{in}} - C_s^{\text{out}}$ has the same physical meaning as the unweighted network centralization C_k . Instead of using the number of counterparts, the advantage of C_s is to indicate the dominating buyer or seller by taking the trade size into account. The two time series of $C_s(t)$ for the two securities are illustrated in Fig. 2(b). Both weighted centralization measures are centered around zero. However, the warrant's C_s has much smaller fluctuations than the stock. In addition, for each security, their C_k and C_s time series evolve quite similarly.

3.3. Assortative index

A network possesses assortative (disassortative) mixing patterns if its high-degree nodes tend to be connected to high-degree (low-degree) nodes. Newman has proposed a measure to quantitatively capture the mixing patterns of directed

Table 1

Summary statistics of trading network metrics for the Baosteel stock (left panel) and its warrant (right panel).

	Baosteel stock					Baosteel warrant				
	C_k	C_s	e_k	e_s	l	C_k	C_s	e_k	e_s	l
Mean	−0.0518	−0.0313	−0.5340	−0.3596	4.7969	−0.0100	−0.0036	−0.3279	−0.1910	6.9542
Median	−0.0498	−0.0284	−0.5212	−0.3476	4.1943	−0.0090	0.0000	−0.3239	−0.1861	6.7217
Maximum	0.9990	0.9145	0.0981	0.8361	29.319	1.0000	0.9456	0.0551	0.2369	18.840
Minimum	−0.9561	−0.8735	−1.0000	−0.9929	1.3077	−0.5453	−0.6942	−1.0000	−0.9998	1.9594
Std. Dev.	0.3106	0.3075	0.1454	0.2179	2.5057	0.0718	0.0801	0.0915	0.0794	1.5639
Skewness	0.1063	0.0581	−0.4336	0.0255	1.9568	3.6731	0.9150	−0.7452	−1.5393	1.2668
Kurtosis	3.1840	2.8411	3.1698	3.6339	9.7857	52.527	20.719	6.0069	16.052	6.8101

networks [14,15]. We simply adopt this measure to see how ask (respectively, bid) nodes are surrounded with bid (ask) nodes, because we also observe that one trader corresponds only to either a seller or a buyer in most of the 5-min trading networks. The assortative index is written as follows:

$$e_k = \frac{\sum_{j=1}^m \frac{ek_j^a ek_j^b}{m} - \left[\sum_{j=1}^m \left(\frac{ek_j^a + ek_j^b}{2m} \right) \right]^2}{\sum_{j=1}^m \left[\frac{(ek_j^a)^2 + (ek_j^b)^2}{2m} \right] - \left[\sum_{j=1}^m \left(\frac{ek_j^a + ek_j^b}{2m} \right) \right]^2}, \quad (5)$$

where ek_j^a and ek_j^b are the degrees of the two endpoints a and b for the edge corresponding to trade j , and m is the number of directed edges from the ask nodes to the bid nodes. As mentioned in Ref. [14], e_k is in the range of $[-1, 1]$ and $e_k < 0$ indicates that the network is disassortative. We also have $e_k = -1$ for star networks with all the peripheral nodes pointing to or departing from the central nodes. The two time series of $e_k(t)$ for the two securities are illustrated in Fig. 2(c).

Taking the trading size into account, we generalize the weighted assortative index through replacing node degree with node strength in Eq. (5):

$$e_s = \frac{\sum_{j=1}^m \frac{es_j^a es_j^b}{m} - \left[\sum_{j=1}^m \left(\frac{es_j^a + es_j^b}{2m} \right) \right]^2}{\sum_{j=1}^m \left[\frac{(es_j^a)^2 + (es_j^b)^2}{2m} \right] - \left[\sum_{j=1}^m \left(\frac{es_j^a + es_j^b}{2m} \right) \right]^2}, \quad (6)$$

where es_j^a and es_j^b are the strengths of the two endpoints a and b for the edge corresponding to trade j , and m is the number of directed edges from the ask nodes to the bid nodes. The two time series of $e_s(t)$ for the two securities are illustrated in Fig. 2(d).

3.4. Shortest path length

The average path length l is defined as the mean value of the shortest path length between two arbitrary nodes in a trading network. We discard the direction and weights of the trading network while computing the path length. The smaller the average path length is, the more tightly the traders are connected in the trading network. The two time series of $l(t)$ for the two securities are illustrated in Fig. 2(e).

3.5. Summary statistics

Table 1 reports the summary statistics of the unweighted network centralization C_k , the weighted network centralization C_s , the unweighted assortative index e_k , the weighted assortative index e_s , and the average path length l of the trading networks for both the stock and its warrant. The Augmented Dickey–Fuller test demonstrates that all the network variables are stationary. The Jarque–Bera test reveals that all network metrics are not normally distributed. The Ljung–Box Q-test for all network metrics rejects the null hypothesis of no autocorrelation. In other words, the time series of these network metrics possess long memory. The fluctuations of the network metrics for the stock are larger than those for the warrant.

The unweighted network centralization C_k and the weighted network centralization C_s on average equals to -0.05 ± 0.31 and -0.03 ± 0.31 for stock trading networks. While for warrant trading networks, we have $C_k = -0.01 \pm 0.07$ and $C_s = 0.00 \pm 0.08$. All the four estimated values are very close to 0, which indicates that both sides of limit order book have very similar order execution patterns. The two maxima of C_k for the stock and its warrant are very close to 1, implying the presence of a dominating buyer in each case. However, the two maxima of C_s are relatively smaller. It means that the trade sizes of the remaining transactions are larger than those in the dominating star structure with a source. In contrast,

Table 2

Summary statistics of financial variables for the Baosteel stock (left panel) and its warrant (right panel).

	Baosteel stock				Baosteel warrant			
	r	v	τ	w	r	v	τ	w
Mean	0.0000	0.0038	6.7599	967173	−0.0005	0.0138	1.1990	11995271
Median	0.0000	0.0025	5.5962	586600	0.0000	0.0096	0.9492	6442832
Maximum	0.0182	0.0291	69.750	19186704	0.1210	0.9520	9.7333	180404593
Minimum	−0.0246	0.0000	0.2407	8350	−0.7492	0.0000	0.0366	51411
Std. Dev.	0.0024	0.0020	4.6957	1226134	0.0149	0.0176	0.9749	14924879
Skewness	0.1078	2.7148	1.7941	4.4427	−15.020	19.285	1.5684	3.0154
Kurtosis	8.3980	19.0289	9.8683	36.3647	693.20	854.47	6.7192	15.8798

the differences of the minima from -1 are relatively larger. We find that these two measures are right-skewed, especially for the warrant. The kurtosis of C_k and C_s are close to 3 for the stock, indicating that the two time series do not have fat tails. On the contrary, the kurtosis for the warrant are very large and hence the two metrics exhibit remarkable fat tails.

We also observe that all the means of the assortative indexes are negative ($e_k = -0.53 \pm 0.15$ and $e_s = -0.36 \pm 0.22$ for stock trading networks and $e_k = -0.33 \pm 0.09$ and $e_s = -0.19 \pm 0.08$ for warrant trading networks), which implies that the trading networks exhibit a disassortative pattern. In addition, the degree of disassortativeness is stronger for the stock than for the warrant. The assortative indexes are left-skewed, except for e_s of the stock. The kurtosis of the two metrics are slightly larger than 3 for the stock and significantly larger than 3 for the warrant. The disassortativeness of complex networks could possibly be explained by maximizing the Shannon entropy [16] and might be related to different price-change phases [17]. However, a simpler explanation is that the trading networks have star-like local structure such that high-degree nodes are more likely to be connected with low-degree nodes.

The average path lengths of the stock trading networks and the warrant trading networks are equal to 4.80 ± 2.51 and 6.95 ± 1.56 , respectively. The shorter average path length of the stock trading networks is caused by the fact that there are more loops in these networks than in the warrant trading networks, which is already evident in Fig. 1. However, the stock trading networks has larger fluctuations in the average path length. We also find that their distributions are right-skewed and exhibit fat tails.

4. Financial variables

We employ four traditional financial variables (return, volatility, intertrade duration, and trading volume) to depict the market behaviors in each windows. All the four variables contain valuable information on the underlying price formation process.

The return r of a security is defined as the logarithmic difference of prices:

$$r(t) = \ln[p(t)/p(t - \Delta t)], \quad (7)$$

$p(t)$ is the price right after the last transaction in window t , $p(t - \Delta t)$ is the price right before the first transaction in the same time window t , and $\Delta t = 5$ min is the window width. The volatility v is defined as the difference between the maximum logarithmic price and the minimum logarithmic price in each time window

$$v(t) = \max_{t' \in (t - \Delta t, t]} \ln[p(t')] - \min_{t' \in (t - \Delta t, t]} \ln[p(t')]. \quad (8)$$

The intertrade duration τ is defined as the mean value of the time elapsed between two consecutive transactions in each time window, whose time unit is second. The trading volume w is defined as the total number of shares both bought and sold during each time window. By performing the Augmented Dickey–Fuller (ADF) tests, the Jarque–Bera tests, and the Ljung–Box Q-tests, we conclude that all these financial variables are stationary, do not obey normal distributions, and possess some autocorrelations, except that the return time series does not exhibit long memory as expected.

Fig. 3 illustrates the time series of all the financial variables defined above for both securities. We observe that the stock fluctuates at a low level while the warrant has much larger fluctuations. The minimum return of the warrant is -0.7492 and the maximum volatility of the warrant is 0.9520 , both in the last 5 min before the expiration time. Moreover, the intertrade durations are very short and the trading volumes increase and are high when the warrant goes to the expiration time. The behavior of the warrant was abnormal, and so were the traders, because the warrant went so deep out of the money that they were almost certain to expire worthless [18]. The plots also show that both return series have evident volatility clustering.

Table 2 reports the summary statistics of the four financial variables. Compared with the stock, one can find that the warrant has larger volatilities, higher trading activities, and larger trading volumes. We also find that the distribution of the stock returns is almost symmetric, while the distribution of the warrant returns is left-skewed. The fact that the kurtosis of the financial variables are greater than 3 indicating that their distributions have fat tails, which are consistent with previous works on stocks and warrants for returns and volatility [19–22], intertrade [23–25], and trading volume [26].

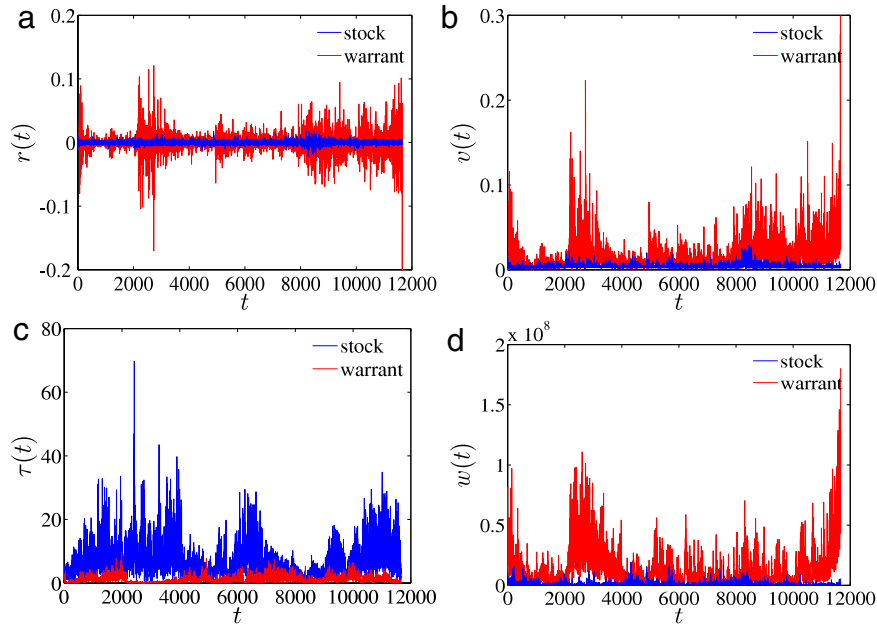


Fig. 3. (Color online) Time series of the four financial variables for the Baosteel stock and its warrant: (a) return, (b) volatility, (c) inter-trade duration, and (d) trading volume. The minimum return of the warrant is -0.7492 in the last window, which is truncated in plot (a) for better visibility. The maximum volatility of the warrant is 0.9520 in the last window, which is also truncated in plot (b) for better visibility.

Table 3

Correlations between financial variables and topological metrics of trading networks.

	Baosteel stock					Baosteel warrant				
	C_k	C_s	e_k	e_s	l	C_k	C_s	e_k	e_s	l
r	0.253***	0.188***	0.037***	0.031***	0.018	0.131***	0.056***	-0.003	-0.017	-0.043***
v	-0.015	-0.012	0.302***	0.051***	-0.119***	0.016	0.003	0.261***	0.072***	-0.095***
τ	0.009	-0.011	-0.382***	0.008	-0.383***	-0.027**	0.007	-0.561***	-0.049	-0.046***
w	0.014	0.035***	0.185***	-0.064***	0.070	0.059***	0.009	0.379***	0.019	-0.183***

*** Significant at 0.1%.

** Significant at 1%.

* Significant at 5%.

5. Correlations between financial variables and trading network metrics

To reveal the connection between the topological structure of trading networks and the behaviors of the stock and its warrant, we present in Fig. 4 the scatter plots between pairs of the five trading network metrics and the four financial variables. For most pairs, the correlation is absent or weak. We estimate the correlation coefficient of each pair between the four financial variables and the five network metrics. The results are reported in Table 3.

The returns of both securities are positively correlated with the centralization measures C_k and C_s at the significance level of 0.1%. A trading network with higher centralization has more dominating buyers and such kind of network structures are resulted from one or a few large market orders, which also leads to positive returns [27,28,22]. Hence, the price is more likely to increase consecutively and results in a large positive return. The stock returns are positively correlated with the assortative indexes e_k and e_s , but uncorrelated with the average path length. In contrast, the warrant returns are negatively correlated with the average path length, but uncorrelated with the assortative indexes e_k and e_s .

For both securities, the volatility is positively correlated with the assortative indexes e_k and e_s , which is in sharp contrast to the negative correlation between volatility and assortative index in E-mini S&P 500 futures market [12], and negatively correlated with the average path length l . More than 99.98% of our trading networks are disassortative, which means that the nodes with larger degrees prefer to connect with the nodes with small degrees. However, by checking the local structures of the nodes with large degrees, we surprisingly find that those nodes having large degrees are connected by multiple edges in some trading networks, known as an abnormal trading motif [9]. Such trading networks may exhibit relatively large assortative indexes, but still negative. Obviously, the abnormal trading activities will increase the market volatilities and adding bridges between hubs will absolutely shorten the average path length in trading networks, which also leads to negative correlations between the market volatility and the average path length in trading networks.

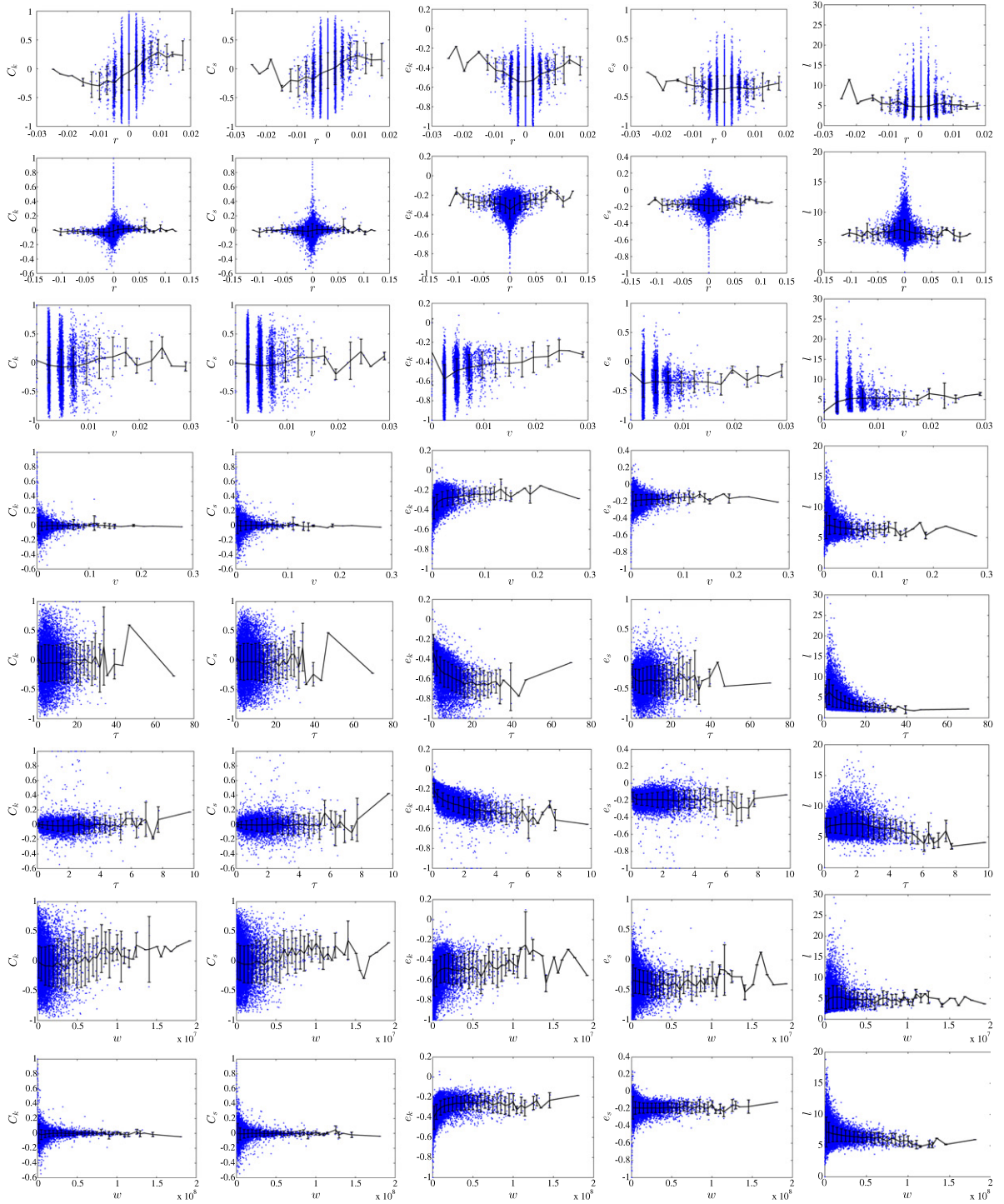


Fig. 4. Scatter plots between pairs of the five trading network metrics and the four financial variables. The lines with error bars are the average values by binning the financial variables. The odd panels are for the stock, while the even panels are for the warrant.

The stock intertrade duration τ is negatively correlated with the assortative index e_k and the average path length l , while the warrant intertrade duration is negatively correlated with the centralization C_k , the assortative indexes e_k and e_s , and the average path length l . Shorter intertrade durations correspond to higher trading activities, which usually happen when there is a lack of dominating buyers and sellers. This could increase the assortative index of trading network. In this situation, the network usually contains a relatively large number of nodes and is less likely to have a star-like structure such that the average path length is more probable to be large.

The trading volume w is positively correlated with the centralization C_s for the stock and with C_k for the warrant. A possible interpretation is the following. Although there is no asymmetry in the volume–return relationship at the transaction level [28], it is well established that it takes larger trading volume to move the price up than to hit it down with the same magnitude of return when the time scale is aggregated (say 5 min as in the present work) [29]. Since a high centralization corresponds to a high return, the trading volume will also be high. The correlations between trading volume and assortative index e_k are positive for both securities. This is because that higher trading volume usually corresponds to large trading activities and the lack of dominating buyers and sellers, which implies higher assortative index as for small inter-trading durations. The correlations between the trading volume and the assortative index are mixed. Moreover, the correlation between the trading volume w and the average path length l is positive for the stock and negative for the warrant. It is hard to give an intuitive explanation for these mixed correlations.

6. Summary

In this work, we investigate the statistical properties of five topological metrics of trading networks and four financial variables in the 5-min moving windows of the stock Baosteel and its warrant. We find that the Chinese security trading networks have disassortative mixing patterns and similar order execution patterns in both sides of the limit order book. The investigated financial variables are stationary and possess fat tails. We also find that, for both stock and warrant, the return, volatility and trading volume positively correlate with the centralization or assortativity, while the intertrade duration is negatively correlated with the network centralization measures and assortative indexes. Our results show that the stock and its warrant share many common topological properties in their trading networks, but also exhibit idiosyncratic characteristics.

Our findings show that topological metrics of trading networks can be used to characterize the temporal properties of financial variables and have the potential to gain a deep understanding of the price formation process from a novel perspective. It is also possible to design possible techniques to identify outliers in the time series of trading networks, which is able to underpin clues for abnormal behaviors in the price dynamics and provide information about possible price manipulations.

Our results are based on non-overlapping moving windows of 5 min. Other window sizes have also been chosen in previous works. For instance, Jiang et al. [2] and Sun et al. [10] use one day in their analysis. There is no objective criterion on the choice of window size. We believe that our results are robust for intraday trading networks. Our motivation is to have longer time series and not to introduce too much noise at smaller windows. In addition, it would be very interesting to follow the idea of Sun et al. in Ref. [10] to check the possible causality between network metrics and financial variables at the intraday levels, which could be done in a future work.

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